

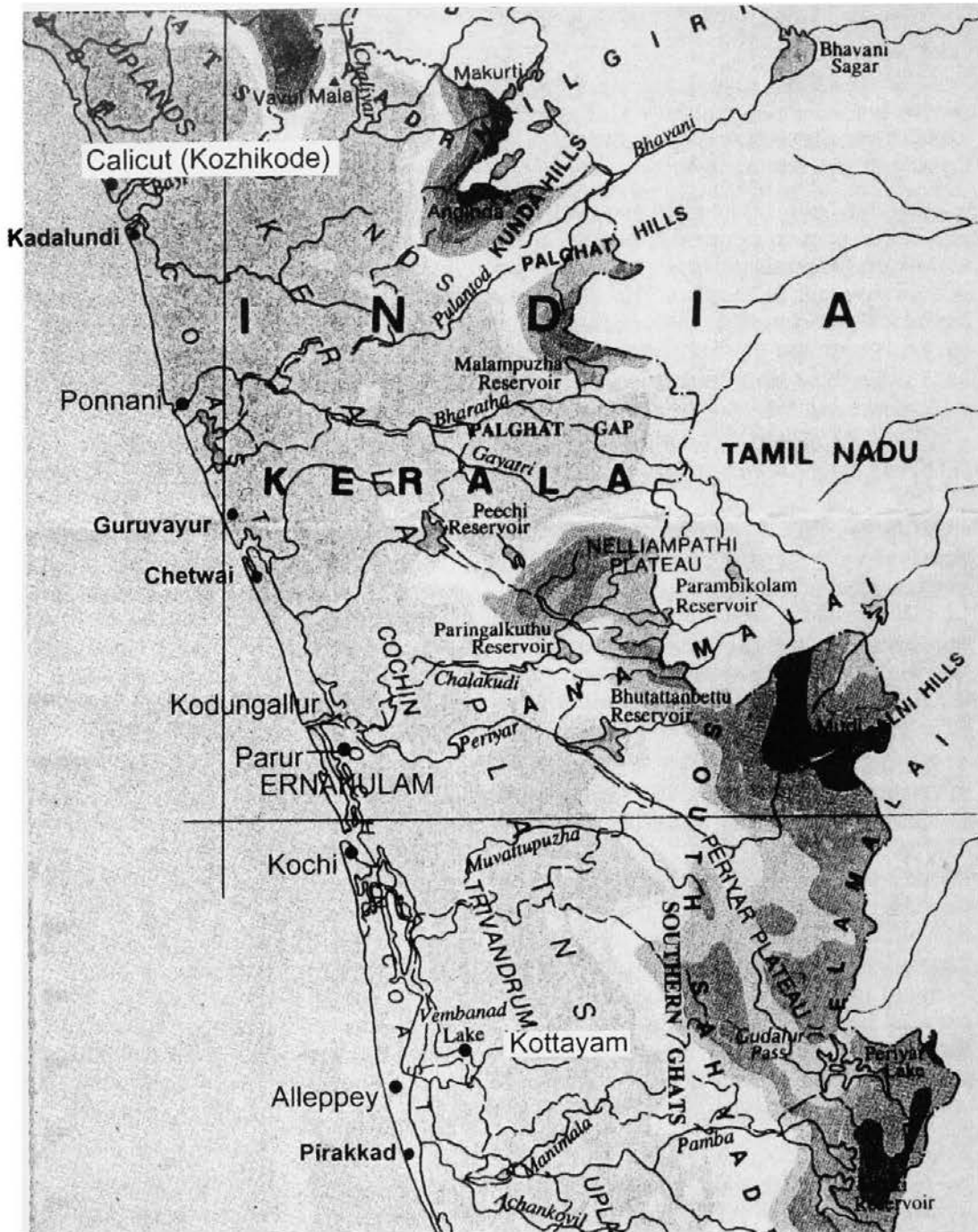


Fig. 1. The coast of Kerala today.

In search of Muziris

Rajan Gurukkal and Dick Whittaker

The importance of Muziris in Roman trade with India does not need any underlining. The port on the Malabar coast of modern Kerala figures prominently in the descriptions of classical geographers, it receives mention in the earliest Tamil poems, and it has come into the news more recently through the publication of a Greek papyrus from Egypt. It is also clear from the amount of Roman silver and gold coin found in S India — which gives some substance to Roman estimates of money haemorrhaging out to India — and from the value of the eastern cargoes recorded coming into Egypt and Rome that the trade was neither casual nor modest. All this is well known and has been carefully studied.¹ The oddity, or pity, is that, despite the many ports listed in ancient authors along some 600 km of the Malabar coast, not a single one has been identified for certain, and not only has produced any serious archaeological evidence of Roman contact. As for Muziris, the most important of them all, we have only a vague idea of where it was located.

Almost every earlier study has placed Muziris at Kodungallur (Cranganore/Cranganur in its Europeanised form) at the mouth of the Periyar river and north of Kerala's main modern port of Kochi (Cochin) (fig. 1). That is reasonable enough. The Periyar is the greatest river in Kerala and runs down from the towering western ghats to the sea. But where exactly on the Periyar? Kodungallur is the name given to a large zone, incorporating a number of small towns of which Kodungallur itself is one, strung out along the road that runs north for several kilometres from the Periyar parallel to the coast and the inland waters of the river Pullut. But how certain is this, anyway? These were the questions we had in mind when a group of us² decided to take a closer look at the evidence, both in the literature and on the ground. Ultimately, only an excavation can answer the questions for certain, but perhaps we could narrow down the options.

The ancient literary sources

The first task was to re-examine the ancient literary evidence. The anonymous Greek pilot who wrote the trader's handbook entitled the *Periplus of the Erythraean Sea* is the earliest and the most precise about geographic directions.³ Muziris, he said, lay between the port of Tyndis to the north and Nelkynda to the south, each stage being about 500 stades (= c.92 km) or roughly a day's run for a ship. But the figure included the fact that both Muziris and Nelkynda were inland river ports, Muziris 3½ km from the coast and Nelkynda some 22 km from the coastal station at Bakare.⁴ A few years later, Pliny the Elder, a Roman naval official whose maritime information about the crossing to India has been proved generally reliable, added that in his day the danger from pirates made Becare, which belonged to a tribe called the Neacyndi, a preferable trading port to Muziris. This was partly because the traders had to lie a long way off shore at Muziris and to be loaded from lighters.⁵ Although Pliny did not explain

1 An excellent guide to recent bibliography is given by De Romanis and Tchernia 1997, reviewed by C. R. Whittaker in *JRA* 13 (2000) 691-92.

2 Including R. Thapar, of the Jawaharlal Nehru Univ., Delhi, the present authors, and various colleagues from universities in Kerala, to whom we offer our thanks.

3 Easily the best edition of the *Periplus Maris Erythraei* (hereafter referred to as *PME*) is by Casson (1989); its date is now more or less accepted as c.A.D. 50, some 10-20 years before Pliny the Elder wrote. The main relevant passage is *PME* 54-56.

4 *PME* 54; the text is damaged but probably reads '(Muziris) lies on a river 500 stades (furlongs) from Tyndis by river and sea, and from [?the river mouth/coast/sea] about 20 stades. Nelkynda is just about 500 stades from Muziris, likewise by river and sea ... It too lies on a river about 120 stades from the sea.'

5 The main information is in Pliny, *NH* 6.104-5: *NH* 6.104 says 'Muziris, the first emporium of India, is not to be sought because of pirates nearby ... Furthermore, the shipping station is a long way from the

why this was necessary, it could have been that silting up of the river mouth made it impossible for a fully loaded ship to cross the sand bar, especially since, as the *PME* said, 'large ships sail to these trading ports due to the volume and quantity of pepper and *mala-bathron*' (a cinnamon leaf). At Becare, Pliny implied, things were different, although he commented that here too small boats made of hollowed tree trunks brought the produce to the port.

Both authors were clear that the principal cargo was pepper, coming for the most part from a region called Kottanarike or Cottanara, which must surely be the modern Kottanadu district centred on the town of Kottayam. We might guess that the ports were reasonably substantial, since corn, the *PME* said, was unloaded for the use of 'those involved in shipping' (οἱ περὶ τὸ ναυκλήριον), though not for the merchants (ἐμπόροι). Presumably, therefore, the shippers and their crews disembarked and lived in the ports while waiting several months for the return monsoon winds, while the trading organisations and warehouses must have been local. Both writers also thought it important for the benefit of traders to add information about the political order of the region that we ascribe to the Cera kings in this period. Pliny talked of a 'king of Muziris' named Caelobothras (called Keprobotos by the *PME*) who lived in the interior while ruling a kingdom extending to Tyndis. The names are obviously Greek rendering of the Sanskrit word *Keralaputra*, found in the Asokan Edicts of the 3rd c. B.C., meaning sons of the Cera clan and referring not to any one ruler but to the descent group or ruling clan as a whole.⁶ Our two writers agree that Becare-Nelkynda was ruled by a king called Pandion, who also lived in the interior — at Modura, said Pliny, which was obviously Madurai, the historic royal city of the Pandyan kings.

One century later Ptolemy, the Alexandrian geographer, added two other coastal sites between Tyndis and Muziris, Bramagara and Kalaikarias, although neither carried the title of 'town' like Tyndis, nor of 'emporium' like Muziris.⁷ But Ptolemy produced three extra, potentially useful pieces of topographic information:

- 1) that on the same latitude as Muziris but inland by 20' was the mouth of a river called 'Pseudostomos' or 'false mouth', false perhaps because it did not open onto the sea;
- 2) that Muziris was by now the only 'emporium' on the Malabar coast, a title not given to Nelcynda, which must have meant that Muziris was the main pepper port in the 2nd c. A.D.;
- 3) that inland at Karoura lay the 'royal seat of Kerobothros' and near it 'Pounata, where beryl is found'. Pounata must surely have been the famous beryl mines near Coimbatore, despite attempts to locate it at Mysore.⁸ Ptolemy's evidence of Muziris's importance was reflected by a contemporary from Alexandria, Lucian, who satirised the bogus claims of historians to veracity 'by a long letter from Muziris' — a name, therefore, familiar to his readers.⁹

The importance of Muziris as a trading centre for Alexandrian shippers in the time of Ptolemy has been confirmed by its mention in a celebrated papyrus now in the Vienna collection (*P. Vindobona* G 40822),¹⁰ which is a written contract dating to c.A.D. 150 for the transport of goods from the Red Sea port to the customs-bonded warehouse at Alexandria. A key phrase in the contract reads 'when the time falls due for the repayment of the loan in the agreement(s) referring to Muziris' (ταῖς κατὰ Μουζεῖριν ... συνγραφαῖς). Unfortunately, exaggerated statements have been made over how much the document tells us about Muziris, since all we can deduce is that some loan agreement(s) [either singular or plural] had been contracted between a

land, and cargoes are brought in and carried out by light boats.' On the accuracy of Pliny, see De Romanis and Tchernia 1997.

6 The transfer of the name of a clan to the king is often found. The termination *-bothra* transliterates *-putra*, which literally means 'son', probably indicating a clan.

7 The main information of Ptolemy is to be found in his *Geog.* 6.1.8-9. The standard edition is still that by McCrindle 1885, reprinted with an introduction by Shastri in 1927.

8 Ptol., *Geog.* 1.1.86; Warmington 1928, 113, McCrindle 1885, *ad loc.*

9 Lucian, *How to write history* 103; for other references by Lucian to India, see Whittaker 1998, 18.

10 Published with translation by Harrauer and Sijpesteijn 1985; cf. Casson 1989, 14.

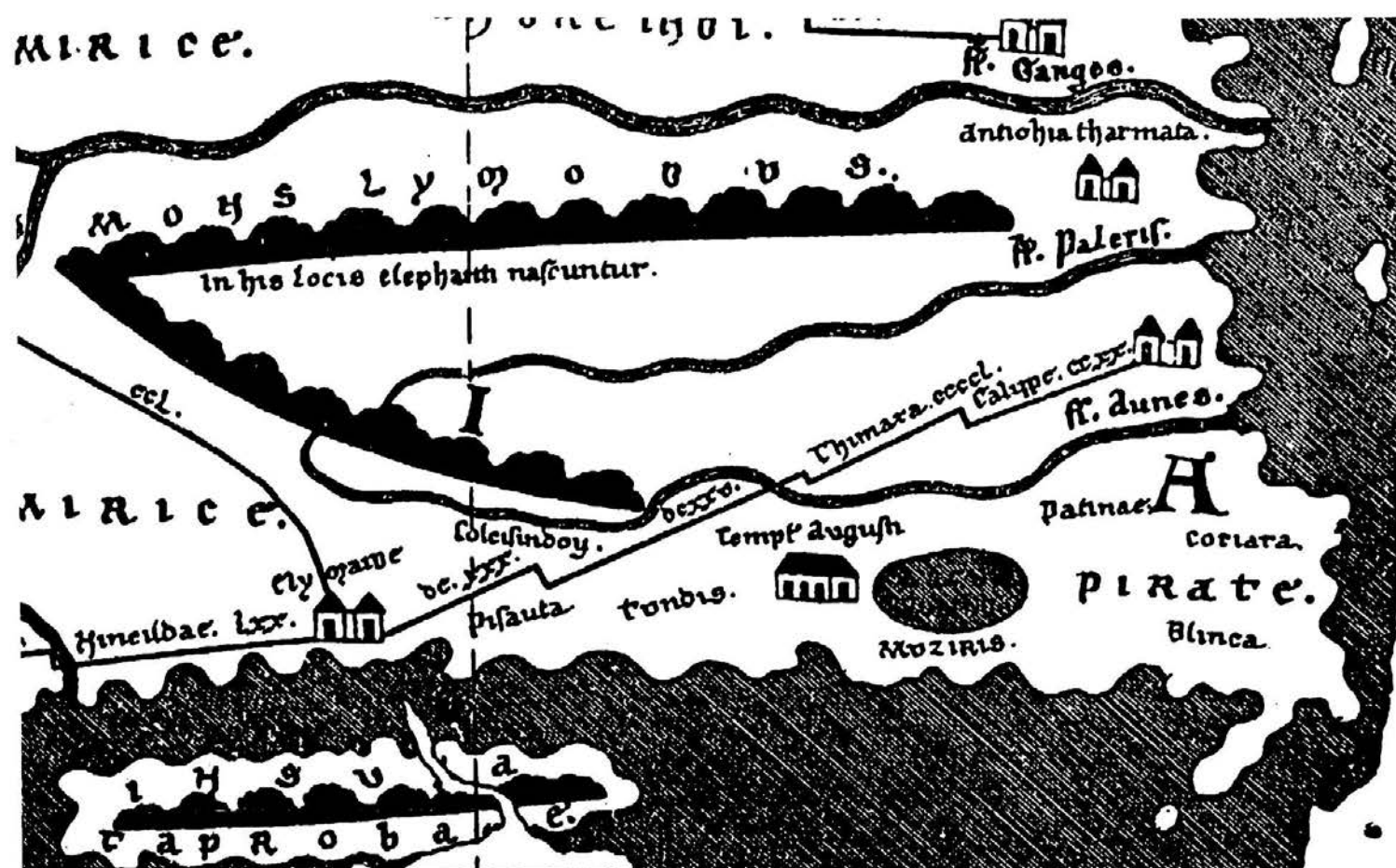


Fig. 2. Section of Peutinger Table.

rich money-lender and a merchant importer. Both parties could have been resident in Egypt, although that is not indicated. What the document does not say, however, is that a contract was made in Muziris, as has been asserted. It therefore tells us nothing about the signatories to the agreements nor where the contracts were drawn up — much less whether there was a Roman agent “resident at Muziris” or whether there was “a trade agreement between a *vanikar* [a Tamil merchant] from Muçiri and a trader from Alexandria” — although all this has been claimed.¹¹

The only clue we have to Greco-Roman foreigners permanently resident in the port comes from the Roman road map, the Peutinger Table (fig. 2) which, though deriving from a Roman original of the 4th c., contains information which could date back to the 2nd c. A.D.¹² Both Muziris and Tondis (Tyndis) are marked there, although neither were apparently on any standard route. Behind the name MUZIRIS on the map there is a large lake, marked LACUS MUZIRIS, beside which is an icon marked *templ(um) Augusti*. The lake can only be a representation of the famous backwaters of Kerala, but the accuracy of the map is not enough for us to be sure that the temple of Augustus was actually at Muziris (even if that seems the most likely possibility) rather than somewhere in the vicinity. Centres for worship of the emperor in the Roman world were usually signs of a resident Roman citizen community, although there is at

11 Casson 1989, 14 and 1990, arguing that *κατὰ Μουζείριον* means “at Muziris” rather than “referring to Muziris”; Champakalakshmi 1996, 121. A recent study of the papyrus by Rathbone 2000 agrees with our translation.

12 The *Tabula Peutingeriana*, now in the National Library at Vienna, derives its name from the 16th-c. scholar Konrad Peutinger who owned it. The existing map is a late mediaeval copy, although deriving from a 4th-c. original, which itself incorporated itineraries going back perhaps to the 2nd c. A.D.: Dilke 1985, 112-20.

least one exception in the cult set up by Arabs in the Hejaz of Saudi Arabia.¹³ At all events, here at least was evidence of the prominence of Muziris in the perception of the Romans, which extended to the 4th c. A.D.

This should warn us against automatically assuming that there was a decline in the importance of Muziris and Roman trade between the 2nd and the late 4th c., as is often done simply because of the decline in the numbers of later Roman coins found in S Indian hoards.¹⁴ In truth, late Roman copper coins of the 4th and 5th c. have been found all over South India, although they have never been properly documented, and more later gold coins have been turning up.¹⁵ Furthermore, in the library of Ambrose, Bishop of Milan in the later 4th c., a strange work about Brahmins, which was translated and adapted into Latin in the 4th c., referred to 'Muziris, the *emporium* of the whole of India this side of the Ganges'. The original Greek informant of the 4th c. made no mention of Muziris but only of a people 'who collect pepper', which the translator naturally assumed to be Muziris.¹⁶

Pepper, too, is being found in increasing quantities in the excavations of warehouses dating to the 4th and 5th c. at the Red Sea port of Berenice, which is a reminder that among the ransom demanded by Alaric the Goth from the city of Rome in 408 was 3,000 lbs of pepper.¹⁷ We assume, too (but cannot prove), that the several references in the 5th and 6th c. to Indian merchant ships in the Egyptian port of Clysma at the head of the Red Sea meant that cargoes of pepper were still arriving in Rome.¹⁸ The Malabar (or Male) coast was actively exporting pepper when Cosmas Indicopleustes drew attention to the fact in the 6th c., although Muziris itself did not figure in his list of ports.¹⁹

Apart from the western evidence about Muziris, Muçiri or Muiri (the Tamil forms of the name) also featured, if not frequently at least prominently, in early Tamil poems which have been collected in the so-called Sangam literature. This is not the place to discuss the problem of dating these tantalising sources, and it must remain uncertain whether it is possible to make any valid comment about the precise historicity of the evidence they contain. The texts are now considered to consist of layer upon layer of information and interpolations deriving from the entire proto-historic Iron Age and Early Historic period between the 3rd c. B.C. and the 3rd c. A.D. The poems were collected, edited and classified into anthologies three or four centuries after their genesis, when they received mediaeval commentaries.²⁰ For the purposes of topography, however, the descriptions of Muçiri are valuable, from whatever date they stem. There are two passages. The first is from the *Akananuru* (149.7-11) an anthology of 400 poems in

13 Whittaker 1994, 127, for references.

14 Warmington 1928, 113, gives a picture of continuing trade, although he believed the decline began at the end of the 2nd c.

15 See the comments by Krishnamuthy 1994, 8, although he subscribes to the orthodox numismatic view that a decline in Roman trade can be traced by the drop in numbers of gold coins after the mid-3rd c. A new hoard of 5th-c. gold Roman coins, found near Madurai, is reported in *The Hindu* of 20.6.98.

16 *Muzirim totius Indiae citra Gangem emporium*. The *De moribus Brachminorum* was ascribed to Ambrose and is preserved under his name in the *Patrologia* (PL 17.1133). The original, written by Palladius, an unknown Greek, claimed to have information from a Theban advocate (*scholasticus*) in the mid-4th c. who, on his way to Sri Lanka, was imprisoned by the king of the 'Bedsades who collect pepper': Derrett 1960, esp. 76-78; Yankovsky 1962.

17 Excavations at Berenice are reported by Sidebotham 1996, 15 and 443, who speaks of "newly booming trade". Alaric's ransom is recorded in Zos. 5.41.4; but we should not exaggerate its importance, since this is only 1½ tons, not much of a cargo for a ship of 400 tons.

18 Petr. Diacon. 116, Ant. Placent., *Itin.* 41.

19 Cosmas Indic. 11.337; McCrindle 1927, 52, identifies two of the 5 named ports, Mangarouth and Poudopatana respectively, as Mangalore north of Muziris and Ptolemy's Podoperoura south of Muziris, perhaps mod. Udayamperur (= Portuguese Diamper). If so, one of two ports in between, Salopatana and Nalopatana, might be a name given to Muziris at this date.

20 Recent studies include Champakalakshmi 1996 chapt. 3; Gurukkal and Varier 1999, esp. 158-60.

the collection called the *Ettuttokai* (containing 8 such anthologies), generally regarded as relatively early. It spoke of 'rich Muçiri' visited by 'Yavana' [Greco-Roman] ships bringing gold for pepper to a port 'dissolving the white surf of the Culli [Periyar] river'. The second is much longer, from the *Purananuṣu* (343.1-11), another anthology of 400 poems in the *Ettuttokai* collection. It described the rice and pepper piled up at Muziris between the ships and the houses, the rice being exchanged for fish and the pepper for gold. The port, it said, was a town noisy with the 'drum of the roaring sea' but 'gold brought by the ships reaches the shore by boats'. An educated guess might put the date of these descriptions to the 2nd c. A.D.; but the later the date, the later the proof of activity at Muziris. The name of the river Culli also crops up in the Mauryan text of Kautilya's *Arthasastra* (2.11.2) as 'Courneyam' a source of pearls, which subsequent Sanskrit and early Malayalam commentaries glossed as a 'river by the name of Curni, near the city of Muraci in the country of Kerala'.²¹ Similar place-names on the bank of the Periyar, such as Curnikkara, have survived to the present day.

The characteristics of Muziris and Karoura

To summarise the ancient written sources, the sort of site one should be looking for is a port, certainly, but a port on a river inlet that silted up a few kilometres from the sea. The sea, however, was still near enough to make its roar heard and to cause tidal surf on the river. Nearby was what looked like a 'false mouth' of the river or the sea. Large ships could not or did not dock at the port itself, implying that there were no deep water jetties or quays, only wharves for light ships in front of the town houses, along which were piled the pepper and rice for loading onto small boats. The wharf was perhaps no more than a bank staked with palm trunks, much as one sees today along the rivers of Kerala.²² Nevertheless, the port was the centre of the pepper trade, and apparently remained so for several centuries. Therefore, it had good access to the hinterland.

Away from the dockside, there must have been a busy settlement, although there is no compelling reason to imagine many brick or stone houses in a region which still predominantly employs wood, cement and mud for building. If foreign traders had spent part of their year in the port, one would have expected warehouses and perhaps even a separate merchant quarter, as has been suggested from the excavations at the Coromandel ports of Arekamedu and Kaveripumpattinam.²³ No doubt red laterite blocks were sometimes cut from age-old alluvial deposits and used as bricks for monumental buildings, much as happens today. The temple of Augustus could have been one such monument, and presumably it occupied a prominent, visible site. But there is not the slightest evidence in the ancient sources that a Roman garrison was stationed in the port, although this allegation is regularly made.²⁴

21 As with so many Indian sources, the reference is difficult to date because of later interpolations. The second book of the *Arthasastra* is considered by Trautmann 1971 to be early, possibly 2nd c. B.C., although the commentaries were obviously later. The commentaries are Bhattaswami's Sanskrit *Pratipadapancika*, the anonymous early Malayalam *Bhashavyakhyana*, and Bhikushu Prabhamati's fragmentary Sanskrit *Canakyaatika*. For all these references and their dating, see Narayan 1993.

22 Wooden stakes were the earliest form of the wharf at Kaveripumpattinam, carbon-dated to c.250 B.C. and associated with Megalithic Black and Red Ware; the stakes were incorporated into a brick platform some three centuries later; Soundara Rajan and Raman 1994, 25.

23 Begley 1996, 15; Champakalakshmi 1996, 127-28; the name Manigramam for one of the early dependent settlements of Kaveripumpattinam is the same as the name for a merchant guild, as known in the Middle Ages.

24 The story originated apparently in Logan 1887, vol. 1, 199, and has been repeated by many others, e.g. Champakalakshmi 1996, 179. A Roman garrison is, of course, quite different from the employment by local rajas of *Yavana* (foreign) mercenaries, who appear in the Tamil poems and epics (*Patirrupattu* 2.10; *Silappadikaram* 14.78).

Apart from the port itself, one might expect the evident prosperity of Muziris over the centuries to preserve some trace of the huge quantities of gold and other coins exported from the West, not to mention any local currency. On the analogy, however, of other ports in Tamil Nadu known to have had contacts with Roman trade — Arikamedu, Kaveripumpattinam or Alagan-kulam (in the Palk Straits) — this could be a vain hope, since the excavations at these sites have yielded only a few, stray or surface coins. Rather, it is around the up-country river sites, such as Tirukkoyilur (for Arikamedu) or Madurai and sites on the upper Kaveri river, the so called “royal centres”, where most hoards have been found.²⁵ The explanation may be that gold and silver had a greater tendency to be melted down or recycled into purchase of foreign goods in commercial ports; but it may also be that the primary function of coins was as high value objects rather than as investment capital. Whatever the reason, the same absence of currency in port areas appears to be true of the considerable numbers of late Roman copper coins. Almost none have been found in the ports. With copper, however, there are additional problems over the site finds, since it was still accepted as circulating specie in the bazaars until recently, and it has been poorly recorded.²⁶ If it is true that counter-marked and slashed pieces in the hoards of both Roman coins and local Karsapanas (of which there are many) were marks of ownership or identity tags for individual banking deposits, it suggests that hoards were often laid down well after the *terminus post quem*, and would be, therefore, of limited guidance in dating or topography.²⁷

All this raises the question of the relationship of Muziris to Ptolemy’s ‘capital’ or ‘royal city’ of the Cera kingdom, Karoura. Was it somewhere near Muziris and therefore likely to provide a clue? From the point of view of the considerable numbers of Roman or local Cera coins and deposits discovered there, there is no doubt that Karur, a city some 70 km west of Tiruchirappalli in Tamilnadu, was a tempting choice. It has been very little investigated, and what small excavations have been made have not been published, as it quickly disappears beneath modern housing developments. There is a Tamil-Brahmi inscription in a near-by cave, sometimes dated to the 1st c. A.D., which contains the names of three generations of Cera kings or rajas, and Karur figures prominently as the Cera king’s residence in the Tamil poems.²⁸ Against this evidence, however, is the problem that Karur lies over 200 km from the Kerala coast as the bird flies, and objections have been raised against the assumption that the Iron Age or Early Historic societies reflected in the Tamil poems had the political or social organisation to form such far-flung kingdoms, since power over lesser chiefs was essentially exercised through predatory raids.²⁹ Pliny, however, noted that Mudurai, the royal city of the Pandyan, did exercise political control over Bekare-Nelcynda, the port just south of Muziris, even though far

25 A useful map and list of coin finds in Tamilakam (though not quite complete) is given by Champakalakshmi 1996, 156-74. A more accurate map of hoards and stray gold coins in Kerala can be found in Sathyamurthy 1992, 31.

26 Krishnamurthy 1994, 9 and 115-16. There are persistent reports under the early British administration of copper coins in use as small change.

27 Suresh (unpublished).

28 We are grateful to S. Suresh for allowing us to see his forthcoming paper on Karur, which provides this information. The Cera chieftains recorded on the inscription at Pugalur, near Karur, are recorded by Mahadevan 1966, nos. 56-57, who now (pers. comm.) dates it to about the 2nd c. A.D. on palaeographic grounds. If the ruler Cellirumporai who is named on the inscription is the same person as Perunceral Irumporai in the Sangam poems, he is normally (though tentatively) dated to the 3rd c. A.D. But this is a vexed question. The literary allusions are discussed by Subrahmanian 1980, 156.

29 Gurukkal and Varier 1999, 202, concede that the core area of the Cera chieftain of the major lineage (*ventar*) was at Karur and that Muziris was a “peripheral strategic point”, while on 193 they call the Cera chieftain of the Pugalur inscription “a chieftain of the collateral branch of the Cera lineage reigning from Karur”. I. Mahadevan (pers. comm.) has indicated his strong disagreement with this view: “The Nanda-Maurya empire ... extended over 2,000 kms by the end of the 4th c. B.C.! The Satavahanas, contemporaries of the Tamil kings ... ruled over a vast territory including Western Maharashtra and the Godavari-Krishna valley in Andhra.”

away in the interior³⁰ — just over 200 km, as it happens. If Madurai why not Karur? But would Pliny or Ptolemy from their Roman imperial perspective have understood the pre-state condition of the Ceran polity, the notion of collateral branches of rulers, or the anachronism of administrative capitals? Karoura (Karur) therefore remains a vexed question.

Searching on the ground

(a) *At the mouth of the Periyar river*

So much for the written documents. A search on the ground would naturally begin with maps, particularly those of Kodungallur/Cranganore, the site traditionally associated with Muziris. Unfortunately, large-scale survey maps do not exist in Kerala. The only detailed maps of the coast available are those of surveys dating from the British colonial administration, which by now are rendered hopelessly out of date by modern roads and urbanisation. Though they contained a fair number of place-names in anglicised spelling, they do not mark enough for systematic study of toponyms; but at least they give the main physical features of the landscape.

The difficulties associated with prospecting along the Kerala coast are obvious, since this is a tropical landscape of water, coconut palms, and rice. Growth is prolific — of population as well as plants. There are vast backwaters of lakes, rivers and canals, with many areas discouragingly marked on the old maps, “usually flooded from June to January”. Clearly some of the prospecting must be done by boat, but access will always be difficult because of the houses and buildings everywhere. Almost every house has its own compound, a plot of ground yielding coconuts, bananas, papayas, and garden produce that grow almost without trying. The Arab traveller Ibn Batuta remarked in the 15th c. that in Kerala ‘There is not a foot of ground which is not cultivated’. And of Cranganur (*sic*), in particular, Simon Grynaeus wrote in 1532, ‘It has no walls around it, and it is thirty miles long because the houses are not contiguous ... but are interspersed with spaces so that it would seem every family lives in a separate village. Through it flow many rivers, and many of the houses are built on the rivers’.³¹ The only difference today appears to be that the spaces have been filled in and many houses look extremely prosperous.³²

On top of this there is the heavy soil erosion and alluvial deposits brought down from the hills by the monsoon rains, which have caused enormous changes to the coast line and silting of rivers. The most cataclysmic change, which every book says has had a profound effect on Kodungallur, was the great cyclone and flood of 1341, which opened up the present harbour of Kochi and the Vembanad backwater lake system to the sea while forming, according to tradition, the *putuveppu* — the new deposit of land called Wypen (Vipen) island off the coast. That event doubtless changed access to the Periyar river, but geologically it was only the most spectacular of the physical changes and new land formation that have been going on from time immemorial. A geophysical survey of the region has established that some 200-300 years ago the sea shore lay along the line of the Tottapalli river, that is, an inland waterway c.3 km east of the present coast (see fig. 3). Some 2000 years ago it lay even further east along the line of the Pullut river about 6½ km inland; since then there have been successive fluvial deposits of sand and carbonaceous (hence dateable) clay to create new strand lines and beach ridges.³³

If Muziris had been situated somewhere here in Roman times the coast at that time would have run some 4-5 km east of its present line. The regular silting up of the river mouth finally forced Kodungallur to cease activity as a port. On a visit to the base of the Indian Navy at

30 Plin., *NH* 6.105: ‘There [i.e., at the port] Pandion was ruling, while his city, called Modura, was far removed in the interior from the port.’

31 *Novis orbis regionum et insularum* (Basle 1532, chapt. 130; quoted by Segal 1993, 15).

32 The result of oil money flowing in from absent members of Kerala’s families employed in the Arab Gulf States.

33 Mathai and Nair 1988. We have marked the shore-line of 300 year B.P. on fig. 3, following the survey findings of Mathai and Nair.

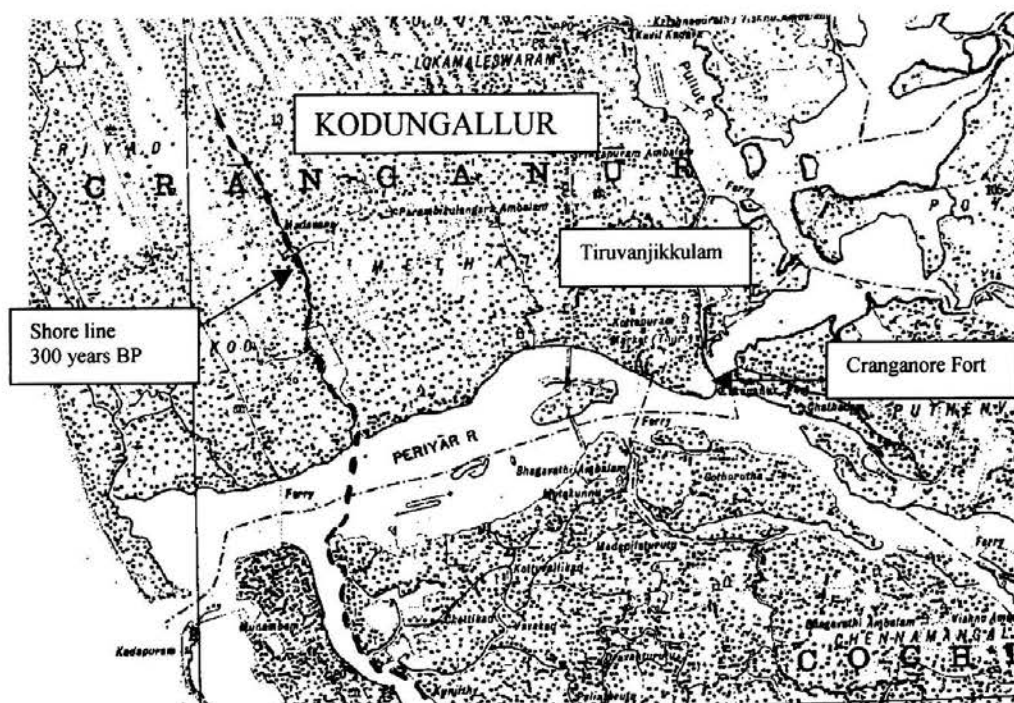


Fig. 3. Map of Kodungallur/Cranganur region (Surveyor General of India 1913, scale 1 inch to 1 mile, with our additions imposed).

Kochi, the naval hydrographer told us that 3 m of silt had to be dredged after each monsoon to keep the harbour open as a deep-sea port. That made it, said the former harbour master, one of the most expensive ports in the world to maintain. The implications of this for archaeology are fairly gloomy, since one can expect everything dating from the Early Historic period to have been buried beneath several centuries of sand and clay.

(b) *To north and south*

Alternatives to the mouth of the Periyar river as the place to locate Muziris have to be considered. To the North, there is the entrance to the backwaters at Chetwai, not far from two Early Historic sites, the temple at Guruvayur and the nearby church at Palayur, one of 7 churches supposedly founded by St. Thomas. But there is no tradition of a port at Chetwai, which, situated as it is some 40 km up the coast, would upset every calculation based on the distances given by the *PME*. To the South, the only real ports today are Kochi harbour, not formed before the 14th c., and Alappuzha (Alleppey), a relatively recent creation of alluvial deposits. Nevertheless, if the Tottapalli river (or somewhere east of it) had in fact been the shore line in the 1st c. A.D., there must have existed some possible entries into the inland river-systems between Kodungallur and Kochi. Sure enough, in the naval museum at Kochi an early Gujarati shipping chart shows three such harbours, at Parur, Konega (Kadamakuti), and Warn[...] (Vaduthula?). Parur lies just south of Kodungallur and adjoins Maleankara, where there was once a Syrian church to mark the traditional landing place of St. Thomas, the other two lie on the delta of a southern branch of the Periyar. By far the most impressive site for Muziris, however, is still at the mouth of the main Periyar river, over 1 km wide at its entrance. Given the available choices, it is difficult to avoid.

(c) *At the confluence of the Periyar and Pullut rivers*

Once that is accepted, there is only one serious contender to look at more closely — somewhere near the confluence of the Periyar and Pullut rivers (fig. 3). The two rivers join about 2 km upstream from the supposed ancient coastline, controlling access to the backwaters or interior, and both were apparently navigable by quite substantial ships — although that is difficult to judge because of the continual silting and dredging that takes place today. The Pullut enters the Periyar from the North at a T-junction, where Kodungallur island (to the west) and the mainland coast (to the east) narrow the entry into the backwater, making the entrance almost invisible from a distance. It was at this point of the island that the Portuguese

built the fort of St Thomas at Cranganore in 1523 — now called Kottamukku, meaning 'fort of the junction'.

As one approaches the narrow river entry by boat from across the broad Periyar, it is difficult to believe that there is a river mouth at all, since it appears as no more than a slight inward dip on the horizon. Could this, then, have been what Ptolemy meant by *Pseudostomos*, 'false mouth'? More interestingly, Muçiri in Tamil means a 'hare-lip' — not a bad description for what one sees where the two banks of the Pullut enter the Periyar.³⁴ The large stretch of water formed by the confluence of the Periyar and the Pullut rivers is today called the Kottamukku/Kuryappilly Kayal. The termination *kayal* means lake, and might possibly have been the lake of Muziris marked on the Peutinger Table. The Portuguese fort was wrecked by Tipu Sultan in 1790 and is now a jumble of stone walls, vines and trees. It occupies the only prominence on the inlet — the sort of place that one might expect a temple of Augustus to have stood. Unfortunately, a detailed study of the surviving place-names and plots in the Kodungallur area has yielded no toponymic clue to the temple. But the site of the fort is crying out for a serious excavation.

Once behind the headland, as one turns north into the Pullut, there is a shallow curved bay to the left (west) created by a protrusion of Kodungallur island into the river, which then recedes after about 3 km to a N-S line, creating another shallow bay. This protrusion of land in the form of a triangle was presumably the result of silting or alluvial deposit and is today criss-crossed with canals and waterways, giving access by the back door, so to speak, to the several villages higher up the ridge — that is, Kottapuram (meaning 'outside the fort'), Tirukkulasekharapuram, Cheramanperambu, Tiruvanjikkulam, and so on — which made up the southern sector of greater Kodungallur. Some of these villages might provide further clues, for it was surely here at this confluence, and perhaps in one of the two bays, that the port of Muziris once stood. Today all we could see was the idyllic, low coast line of palm trees, Chinese fishing nets, black stakes in the water, and the long, gondola-like skiffs which the locals punted around.

(d) Archaeology and place-names

There have been attempts to discover traces of Muziris by archaeology. Two hoards of Roman coins have been found in the administrative district of Ernakulam, of which Kodungallur is a part, one at Panangad (Kumbalam), another more recently at Valluvally (Parur). Valluvally lies on a branch of the Periyar, called Ceriyappilly, which separates off at Aluva (Alwey), only c.13 km from Kodungallur — in other words, close enough to Muziris to make it probable that the coins had arrived through the port. Both hoards, as they are catalogued today, terminated with Antonine imperial issues, but since only 252 coins survive out of the 1,000 or so reportedly discovered at Valluvally, it is impossible to be certain when the hoard was sealed. Roman coins have also been reported from Kodungallur itself, although no one seems to have any knowledge of them.³⁵

To the north of the Kodungallur complex, at various sites along the Pullut river, Megalithic (that is, Iron Age) burial goods have been found which must have belonged to the people who were living hereabouts when the Roman ships arrived.³⁶ Still further north, at Madilakam

34 Muçiri in Malayalam means a mouth with three (*mu*) lips (*ciri*), i.e., with the upper lip split into two. If one looks from Kottamukku 'lake' — that is, from what was once the sea — to the point where the river breaks the horizon, it looks like a mouth with three lips.

35 Induchudan 1970, 171, refers to a report in the *Bulletin of the Ramavarma Institute* 8.2, 99, by T. K. Krishna Menon, claiming that the coins were in the Madras Museum. An old man in the village of Tiruvanjikkulam claimed that, when he was a child, gold coins had been found near the mediaeval site, but were removed by government officials.

36 For example, at Kotaparampu (for which see below) and Panangad, as well as along the route between the Periyar river and the Palghat Gap, which leads to Coimbatore — places where one might expect beryl to have passed on its way to Muziris for export. But no site has been excavated in the Periyar region. See R. Chedambath, *Investigations into the Megalithic and Early Historic periods of the Periyar*

village, excavations have revealed a laterite wall about half a mile long and 6 ft high, with associated Chinese Celadon Ware and coins of the Cola kings of the 10th-11th c. In the southern villages of Kodungallur, excavations at Tiruvanjikkulam, Cheramanperambu and Tirukulasekharapuram produced nothing earlier than mediaeval remains of the 9th-11th c. (the period of the second Cera kingdom).³⁷ There is a persistent tradition that hereabouts stood the capital of the Cera kings, and attempts had been made to associate the excavations with descriptions in the Tamil mediaeval epics. The conclusion from excavations, therefore, is that they say something about the people who lived in this area both before and after the Romans, but nothing of the Romans themselves, apart from the coins they brought.

Place-names, though slippery as historical evidence, might be of some help. There is, for example, a site called Kotaparampu, about 3½ km upstream from the mouth of the Pullut, where Megalithic burials had been found. The prefix *kota-* is a frequent title of early Cera kings, and *-parampu* (meaning 'house') appears again in the place-name of the mediaeval Cera capital Ceraman-parambu. It looked, therefore, like a plausible seat of the Ceras in the 1st c. A.D. and fits the impression given by Pliny and the *PME* that in the 1st c. A.D. the Cera king lived near Muziris.³⁸ If this were so, presumably the Ceras had established a political centre near the coast as a result of the flourishing transmarine trade with Rome and others. A proper archaeological survey at this site might yield results.

There appears, however, to have been yet another early Cera royal centre nearby at Kulur, probably the site called Kulur in the early Sangam collection, the *Akananuru*,³⁹ for which there is also some evidence from epigraphy and the discovery of Megalithic tombs.⁴⁰ Quite close to Kalur is a place called Karur, about 10 km east of Kottamukku, the hypothetical lake of Muziris. Could this, then, have been the Karoura which Ptolemy, a century after Pliny, had recorded as the centre of the Ceras? If it is debatable that the Karur 200 km away in Tamilnadu was Ptolemy's Karoura, this site of Kalur/Karur has the advantage of being much nearer Muziris. It also seems to have been a site of some importance, to judge by the place-names around about. Nearby, for example, is a site called Karurppatanna ('the pan of Karur') about 3 km south of Kottamukku. An old (1818) map of Cochin shows several villages in the area with evocative names, such as Karurvatakku (=north Karur), Karurtheckku ('south Karur'), and Karurmuri. Megalithic tombs have been discovered in the Karur/Kalur region, although none has been excavated. Nevertheless, even if the chronological relationship between Kotaparampu and Karur remained unclear, it is not difficult to imagine that in the political conditions of the 1st and 2nd c. A.D. the Cera kings had more than one political centre of power diffused through collateral families.

Later sources on Cranganor and Shingly

So far we had arrived at the same conclusion as others had suggested, though by adding more precision. There might, however, be more to be learned by examining the fate of Muziris after Roman and Tamil records stopped. The best evidence for continuity of the name Muziris

and Ponnani river basins of Kerala (PhD thesis, Pune Univ. 1997). For maps see Gurukkal and Varier 1999, 107 and 110.

37 Reports are in *Indian Archaeology* 1968/69 and 1969/70; a summary is attempted by Induchan 1970, with references to earlier work.

38 Certain points on the Pullut river near Kotaparampu are still referred to as *kappachal*, meaning 'ship-tracks'. Just north of Kotaparambu is a place called Nattika, which could be Nitriae, near enough to Muziris, according to Pliny, *NH* 6.104, for pirates to have frightened ships away from the port. Ptolemy 7.1.7, however, places Nitriai north of Tyndis, perhaps at Mangalore and, therefore, beyond Cera power; see Casson 1989, 298-99 for references.

39 *Akan.* 168.4-7 speaks of Kulur as the seat of Utiyan Ceral, first among the Ceras.

40 The inscription, dating from about the 12th c. A.D., was found at the temple of Kulur, about 13 km east of Kodungallur; it refers to Kulur as Tirukkulumur (where *tiru-* is an honorific prefix); see *Ramavarma Research Institute Bulletin* 8.1 (1939) 48.

lies in the famous copper-plate inscription, now kept in the Paradesi synagogue at Kochi and thought to date from about A.D. 1000.⁴¹ It is the record of various aristocratic privileges, plus the tolls on ships and carts and land taxes, granted to a Jewish leader (called a 'prince') by the local king, Bhaskararavi Varman, who himself was living at Muyricote.⁴² Muricote is thought in its Tamil form to be Muyiri-kode, meaning 'the landscape jutting out into the sea at Muziris'. The jutting-out point was not necessarily, of course, the site of the port. But given that the grant was made to a Jewish trading guild (*anjuvannam*), the land must have been near the port. This land-grant has generally been linked to the Jewish community that was later extensively recorded at a place called Shingly, Cyngalin or Jinkala, a name which in one form or another occurs in mediaeval Arab and Christian writers from the 12th c. onwards. Shingly has even been claimed to be one of the roots of the name Tirurvan — *jikkulam*, the supposed seat of the mediaeval Cera kings,⁴³ although this derivation is now rejected by Tamil scholars. At Tiruvanjikkulam, one of the villages of Kodungallur, a large artificial pool or tank, traditionally associated with the Cera king, Ceraman Perumbal, is still called Joodakkulam, the Jewish tank.⁴⁴ That makes a strong argument in favour of the idea that this particular sector of Kodungallur was part of the ancient site of Muziris. But whether plausible or not, Shingly was certainly known to Rabbi Benjamin of Tudela as a port in the 12th c.

The name Kodungallur (Cranganore/Cranganur) however, must have had nearly as long a history as Shingly, since the names seem to have been regarded as alternatives quite early on. In the 16th c., Assemani's *Dissertation on the Syrian Nestorians* stated that 'Sciglia is another name for Chrongalor, the city which we say belongs to the people of Cranganore in Malabar'.⁴⁵ But before this a Syrian Christian trading corporation (*manigrammam*) had, like the Jews, recorded grants from the local ruler on another set of copper plate, now preserved in the old seminary of Kottayam, which probably also dated from the 9th or 10th c.⁴⁶ According to the inscription, the Christian 'lord of the town', Iravi Kortalan, was given rights in the port over 'the brokerage and customs due on all ... that may be counted or carried, or whatever it might be, namely *within the river mouth of Codangulor*'.⁴⁷ Those last words showed that shipping was certainly entering the river Periyar to reach the port.

41 The date of the plates has been placed anywhere between the 6th and 11th c., but more recent opinion seems to have settled on c.1000: Brown 1982, 90; Segal 1993, 8-9.

42 The name on the plate appears as *parkaran iravai vanmar*, a Malayalam translation of the Sanskrit *bhaskaran ravi varman*, which is the form that usually appears in textbooks.

43 Yule and Burnell 1903, q.v. "Shinkali", gives references, citing Dr. Gundert in the *Madras Journal* 1844, 120, for the derivation. The various forms of the name that appear in the texts are Shinkali, Shingala, Gingaleh, Chinkali and Zinglanz.

44 Illustrated in Segal 1993, pl. 3; Segal also claims that the name Joodakunnu = Jewish hill was still the name of a site in Tiruvanjikkulam. But other sites with the names Joodakunnu and Joodakulam were pointed out to us near Parur on the S bank of the Periyar. Cf. Brown 1982 for the names at other places, such as Palayur and Vadakkat. There were many Jewish communities in Malabar, as many as 70,000-80,000 Jews, according to Moses de Paiva, *Noticias dos Judeos de Cochim* (1686). The Jews of Cranganore were expelled to Parur and other villages on several occasions in the 16th c.

45 Assemani (1510) cites Jacob, Syrian Bishop of India, for the opinion that *Sciglia alias et Chrongalor vocatur, ea quam Cranganorium dicimus Malabariae urbem*. The Arab chronicler Rashiduddin wrote about 1300 that 'after Manjarur [Mangalore] ... then Chinkali, then Kulan [Quilon]', and a similar list of ports occurs in the Arab writer Shemseddin Dimishqui a few years later: Yule and Burnell 1903, *loc. cit.* But the name does not appear in the Genizah archives, which lists the ports of Kerala as Fandarayna (north of Calicut), Cannanore, and Kolam (Quilon); however, these ports were described mainly for the reception one could expect from the local king, which might not have been a problem in a port where there was already a Jewish sector; cf. Goitein 1987, 453 and 460.

46 The terms *anjuvannam* and *manigrammam* seem to have been used for long-distance trading organizations; the former more specifically for foreign merchants. At the Tamilnadu port of Kaverippumpattinam, the *manigrammam* occupied a quarter in the city: Champakalakshmi 1996, 48-49 and 127.

47 Translation by Mingana 1926, 487 (slightly emended).

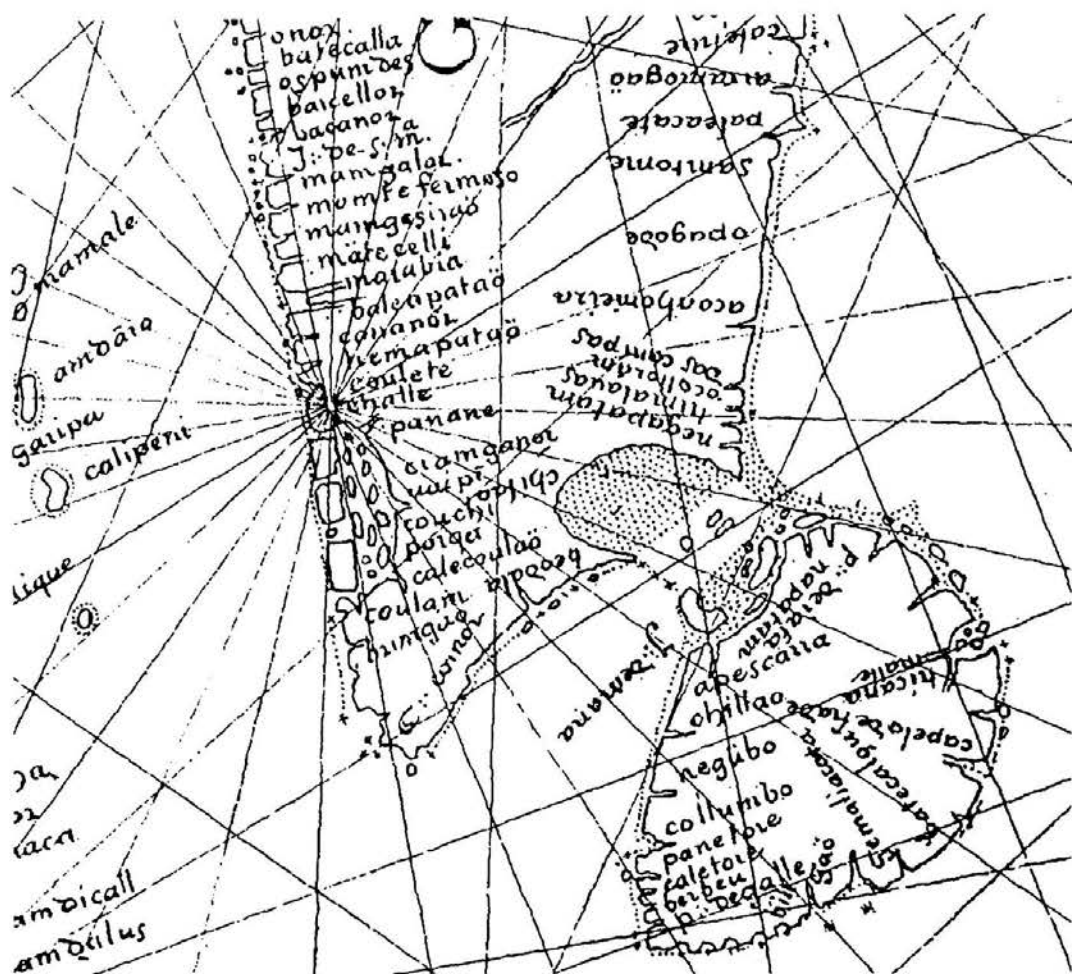


Fig. 4. Part of the Portulan map of Fernão Vaz Dorado, c.1570.

A Syrian lectionary of St. Paul's *Epistles*, dated to 1301, completes the link with Shingly when it says, 'This holy book was written in the royal, renowned and famous city of Shingala in Malabar ... in the church of illustrious martyr, Mar Cyrianus'.⁴⁸ Alongside the Jews, the Nestorian or Syrian Christians, who claimed St. Thomas as their founder, were a prominent presence at Kodungallur when the Portuguese arrived. The anonymous author of Vasco da Gama's first voyage to Malabar, who wrote the so-called *Roteiro* in the early 16th c., said, 'Quorangoliz [Cranganore] is a Christian country and the king is a Christian ... (who) can muster 40,000 fighting men'. And, he added, 'There is much pepper' [to which he then gave a market price cheaper than at Calicut]. When the Portuguese finally attacked Kodungallur in 1505, Jewish and Muslim houses were sacked but Christians were spared.

What these references show is that, contrary to the assertions of modern historians and guide-books, Cranganor (to use its preferred Portuguese form) continued as a prominent trading port long after Kochi became a port and centre of political power. It was for this reason that in 1508 the Portuguese Viceroy, Almeida, wrote to the king recommending a fort at Cranganor to prevent the passage of pepper to Calicut;⁴⁹ and in 1518 Duarte Barbosa named Cranganor as a key town between the kingdoms of Calicut and Cochin, whose river [the Periyar] marked the dividing line.⁵⁰ The Thomas Christians, as the Syrian Christians are called, played an impor-

48 Mingana 1926, 501; cf. Brown 1982, 89-90.

49 Correa 1858, vol. 1, 906, cites Almeida's letter: 'Quanto ao rio de Cochym já eserui a Vossa Alteza que em Cranganor seria bom hum castello forte, em huma tramessa de hum rio que vai pera Calicut, porque lhe tolherá que non passe pera lá alqueire de pimenta.'

50 Dames 1918, *The Book of Duarte Barbosa* vol. 2, 88-89.

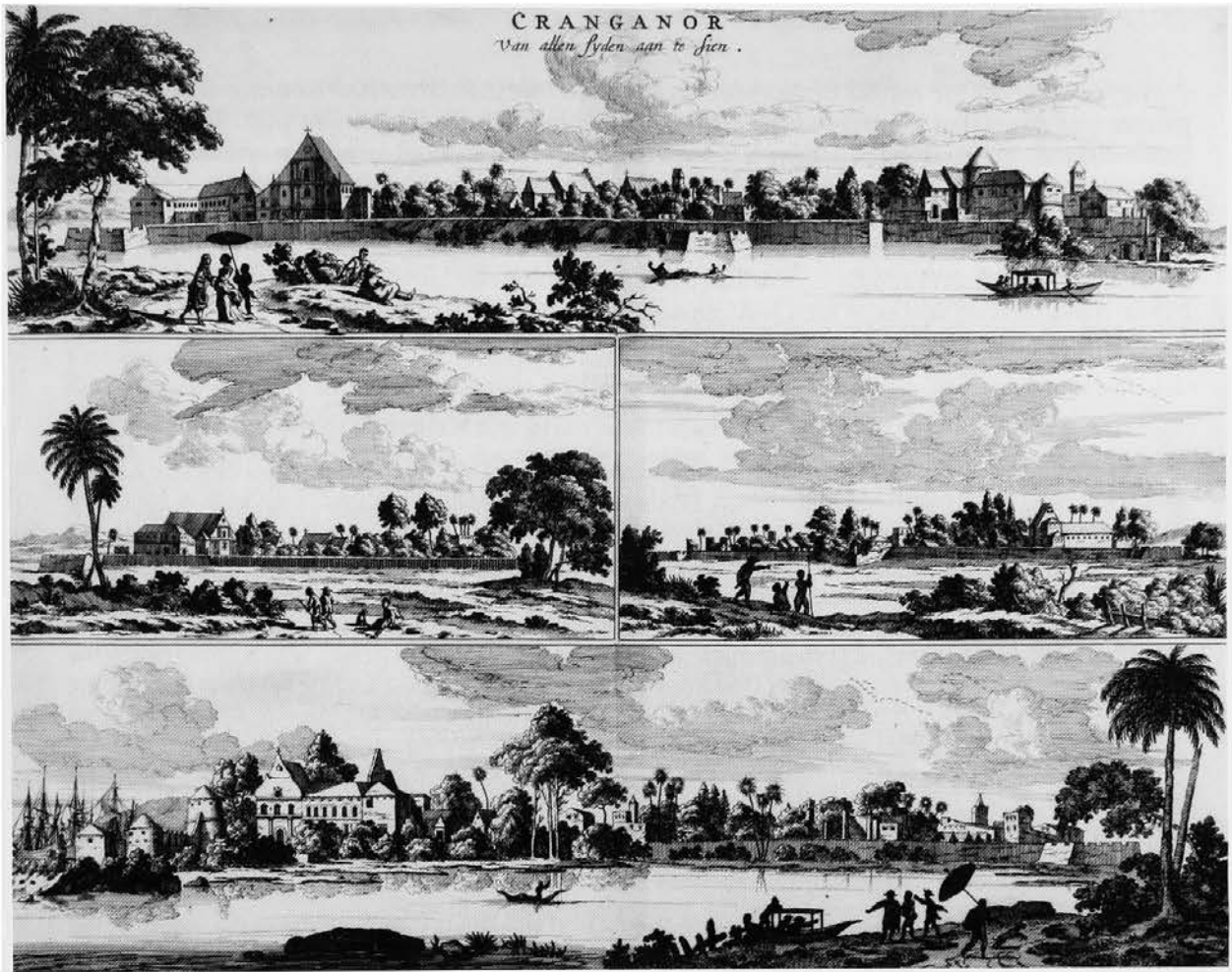


Fig. 5. Baldaeus's illustration of Cranganor.

tant rôle in the town through their control of the pepper trade that so much interested the Portuguese.⁵¹ On the celebrated Portulan map of c.1570 drawn by Fernão Vaz Dorado, now in the British Museum (fig. 4), "Ciamganor" still figures as a port, and it is also clearly linked via the backwaters to Kochi ("Couchi" on the map), despite the formation of Wypen and other islands in the cyclone of 1341.

What we would really like to know is what the site of Cranganor looked like before the Portuguese finally built their fort there in 1523, since so much seems to be pointing to this prominence as a clue to the forgotten history of Muziris. Although this may be a vain hope, a remarkable set of 4 line-drawings published by the Dutch historian, Philip Baldaeus, in 1671, shows Cranganor's fort and port at the time of its capture by the Dutch (fig. 5). In the bottom panel the masts of deep sea ships can be seen docked beside the fort where it faced outwards onto the Periyar river, while on the side facing the Pullut river only lighter boats are pictured. Inside the walls of the town you glimpse from all sides the fort, the 7 parish churches, the Jesuit college and library, and the Franciscan cathedral. At the time, Baldaeus said, the 'king' of Cranganor kept his residence not far away (perhaps at Tiruvanjikkulam), and the Thomas Christians had their College of Chanotte, with a church and school, outside the walls of the town.⁵² Nothing of all this remains except a few crumbling bricks of the fort.

51 Brown 1982, 15. Ferroli 1939, 110, cites a letter of 1523 from the Syrian bishop, Mar Jacob, to the Pope telling him how he had met the Portuguese at Cochín where, he said, he had persuaded his flock 'never more to sell the pepper to the Moors [Muslims] and to bring it clean and dry to your factory.'

52 Baldaeus 1703 (transl. Churchill and Churchill) 629-30. Mathew 1988, 168-69, gives the dimensions of

To some extent the results of this investigation are less than dramatic. Kodungallur or Cranganore was suggested long ago as the most plausible location for the ancient port. That seems to be confirmed. But Kodungallur covers a large area: now the choices must be narrowed down to the bays just beyond the junction of the Periyar and Pullut rivers. Place-names, oral tradition and historical texts of later periods show the value of further study along these lines. Clearly the bays beyond the confluence and the Portuguese fort are significant points which need further investigation through excavation and survey. We can only hope that this will happen before the combined forces of man and climate destroy whatever remains.

APPENDIX

Searching for the ports of Tyndis to the north and Nelkynda to the south

As a footnote to the search for Muziris — if Muziris was indeed at Kodungallur — we describe a rapid look at the ports to the North and to the South, Tyndis and Becare/Nelkynda, respectively, each of which the *PME* said was 92 km distant (which included river-travel).

To the North there are only two real choices; Kadalundi, about 117 km north from Kodallungur, and Ponnani, about 74 km.⁵³ Kadalundi, at the mouth of the Chaliyam river, is *prima facie* the more attractive choice for the site of ancient Tyndis or Tundis, since the place name Kadalundi is supposedly a compound of *kadal*, 'sea', and *tundi*, 'navel'. It is also thought to have been the location of Tondi, which appeared in the mediaeval Tamil epic, the *Silappadikaram* (14.136), as a town in possession of a trading fleet that paid tribute to the Cola king. Today it is a small village, but with the rare distinction for the Malabar coast of an open harbour entrance through a channel 40 ft deep that does not need dredging — historically deep enough for large Portuguese vessels. It was from this port that most vessels carrying Malabar pepper sailed, a factor which convinced the Portuguese that they should build a fort here in 1531.⁵⁴ The village called Nagaram, 'trading place or town', stands on raised ground of original red laterite (not the usual alluvial deposit) at the S entrance to the harbour. Just inside the harbour a villager pointed out to us where a Megalithic pot had been found in the foundations recently dug for a new house on the shore.

Ponnani is a small, bustling port with its own fishing fleet and fish market standing at the mouth of the Ponnani river. The river is a broad inland waterway which runs directly from the Palghat Gap, the natural passage through the high Western Ghats, on to Coimbatore and the Coromandel ports of Tamil Nadu. This route through the Palghat Gap is marked on the Peutinger Table, although it apparently by-passed 'Tondis'. The mines near Coimbatore must certainly have used this ancient route to export their beryl, which no doubt accounts for the number of Roman silver and gold coins found in the vicinity of the mines. In recent historic times, the port used to import salt from the East and to export timber, pepper and other spices. About 6 km up-river from the town of Ponnani, and 1 km from the present river bed, there was a village called Tindilam, not far from a huge paleo-channel of an inland water, now controlled for the paddy fields by a dam. Beside this village was another called Neithallur, whose name had the meaning of 'coastal'. It was not hard to imagine that here was another example of coastal transformation caused by the all-too-evident action of alluvial deposits. Ponnani's port is plagued by sand banks and silting.

Each port thus has a reasonable case to be considered the ancient Tyndis, and there seems no sure way of deciding between the two, short of excavation, since neither fits precisely the distance from Muziris provided by the *Periplus*.⁵⁵ On balance, Kadalundi is perhaps the more probable.

Turning to the South, the coastal changes in the formation of the Vembanad Lake backwaters have been so profound that it is almost anyone's guess where a plausible port of Becare might have been situated, or how it would have linked with the inland town of Nelkynda. The most obvious port today, Allepey (Alappuzha), is entirely the result of recent geological deposits which have created a new spit of land c.3-4 km in width between the sea and the Vembanad waters. That distance more or less marked the line of the Pambiyar (Pampayar) river from the coast, a navigable river for much of its distance eastwards until it reached Chengannur in the foothills. Not far from Chengannur is Niranom, the site of one of the churches supposedly

the fort proper as 600 fathoms (3600 ft.) in circumference and 1½ fathoms (9 ft) in height.

53 Logan 1887, vol. 1, 69 ff. surveys all the ports but all others are too small.

54 Mathew 1988, 188. The fort was a thorn in the side of the Zamorin and was destroyed in 1571, forcing the Portuguese to attempt to build a fort at Ponnani (never completed).

55 Casson 1989, 297, concluded that the odds slightly favoured Ponnani.

founded by St. Thomas; no doubt for this reason it had been thought a possible site for Nelcynda. Otherwise, there is not much to recommend the site, which does not even stand on the river.

Much more plausible seems Kottayam, a large urban centre today and a focal point of the Syrian church in the Middle Ages, half-encircled by the river Menacil which flows into the Vembanad system. Not only were mediaeval copper-plate records that concern the early Christian trading community of Quilon to the south found at Kottayam, but it has always been an important trade centre that linked the eastern pepper and spice-rich hills with the Vembanad waterways. But a single worn gold *aureus* of Nero is its only direct link with antiquity.⁵⁶ Ptolemy gave the name of the river on which Nelkynda stood as Baris, but, despite the fact that the *PME* appears to suggest that Becare and Nelcynda stood on the same river, the huge changes in the Vembanad waterways make it impossible to base much upon this evidence.⁵⁷ A disused canal linking Kottayam to the Vembanad in a district called Vaiaskara bears some resemblance to the ancient name Becare, which could possibly indicate that the coastal port transferred its name to a suburb of Kottayam.

If Kottayam was indeed Nelkynda, the distances provided by the *PME* from Nelkynda to Becare on the coast (wherever it was) and to Muziris (Kodallungur) coincide pretty well. From Kottayam to the coast at Alappuzha, as the ferry goes, is 19 km, against the 22 km from Nelkynda to Becare recorded in the Greek source. The river on which Becare supposedly stood could have been an opening into the Vembanad Lake, since the coast line appears on maps of the 16th and 17th c. as a string of islands (see, for example, fig. 4). Today the main access to the lake is via the Pambiyar (Pampayar) river, which flowed parallel to the coast about 3-4 km inland, but which formerly must have been part of the inland sea waters.

On early maps Becare itself was often identified with Pirakkad (Porakkad) or Porga, today nothing but a sandy beach. Coastal aggression makes it impossible to guess how far inland the site might have been, but memories of a lost harbour are reflected in place-names and legends.⁵⁸ According to the *PME*, the distance from Muziris to Becare was c.70 km (i.e., 92 km minus the 22-km inland river journey), while Kodungallur to Pirakkad measures about 96 km. That could mean that Becare was further north at a site, now lost, which gave access to the Vembanad lake. A Dutch map of the coast in the 17th c. shows the Vembanad waters open to the sea at several places (S. Iago, S. Andre, H. de Padre), although Porca (Pirakkad) is there marked as the principal port between Cotchyn (Kochi) and Coulan (Quilon).

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56 The stock register number of the coin in the museum is given as M/91/487/1, but we were not able to verify this or discover anything about its provenance.

57 Casson 1989, 298.

58 The present group of fishing houses, for instance, are called Devaswam Puthuval, meaning 'recently created temple compound'; down the coast is a point called Punthala, meaning 'new headland'.

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